KCET-2015 (Mathematics)

1. $f(x)=\frac{1}{2}-\tan\left(\frac{\pi x}{2}\right)-1<x<1$ and $g(x)=\sqrt{(3+4x-4x^2)}$

   Find domain of (f +x)

   a. $\left[\frac{-1}{2},1\right)$
   b. $\left(\frac{-1}{2},1\right]$
   c. $\left[-\frac{1}{2},\frac{3}{2}\right]$
   d. (–1, 1)

2. Write the set builder form A = {–1, 1}
   a. A = {x : x is a real number}
   b. A = {x : x is an integer}
   c. A = {x : x is a root of the equation $x^2 = 1$}
   d. A = {x : x is a root of the equation $x^2 + 1 = 0$}

3. If the operation $\oplus$ is defined by $a \oplus b = a^2 + b^2$ for all real number 'a' and 'b', then $(2 \oplus 3) \oplus 4$ = _____
   a. 181
   b. 182
   c. 184
   d. 185

4. If $z=\frac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}$, then $|z|$ is equal to
   a. 0
   b. 1
   c. 2
   d. 3

5. If α and β are the roots of $x^2 - ax + b^2 = 0$, then $a^2 + b^2 = 0$, then $a^2 + b^2$ is equal to _____
   a. $a^2 - 2b^2$
   b. $2a^2 - b^2$
   c. $a^2 - b^2$
   d. $a^2 + b^2$

6. If the 2nd and 5th term of G.P. are 24 and 3 respectively, then the sum of 1st six terms is _____
   a. $\frac{189}{2}$
   b. $\frac{189}{5}$
   c. $\frac{179}{2}$
   d. $\frac{2}{189}$

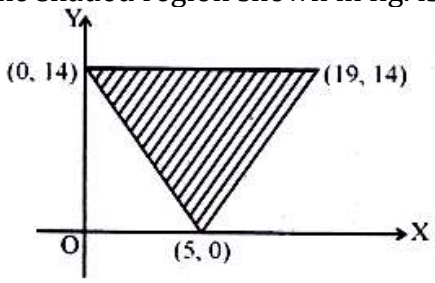
7. The middle term of expansion of $\left(\frac{10}{x}+\frac{x}{10}\right)^{10}$
   a. ${}^7C_5$
   b. ${}^8C_5$
   c. ${}^9C_5$
   d. ${}^{10}C_5$

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8. If $\begin{vmatrix} 2a & x_1 & y_1 \\ 2b & x_2 & y_2 \\ 2c & x_3 & y_3 \end{vmatrix} = \frac{abc}{2} \neq 0$, then the area of the triangle whose vertices are $\left(\frac{x_1}{a}, \frac{y_1}{a}\right)$, $\left(\frac{x_1}{b}, \frac{y_2}{b}\right)$, $\left(\frac{x_3}{c}, \frac{y_3}{c}\right)$ is

a. $\frac{1}{4}abc$
b. $\frac{1}{8}abc$
c. $\frac{1}{4}$
d. $\frac{1}{8}$

9. The shaded region shown in fig. is given by the inequation



a. $14x + 5y \geq 70$ $y \leq 14$ and $x - y \leq 5$
b. $14x + 5y \geq 70$ $y \leq 14$ and $x - y \geq 5$
c. $14x + 5y \leq 70$ $y \leq 14$ and $x - y \geq 5$
d. $14x + 5y \geq 70$ $y \geq 14$ and $x - y \geq 5$

10. $\sim[(-p) \wedge q]$ is logically Equivalent to

a. $P \vee (\sim q)$
b. $P \wedge (\sim q)$
c. $\sim[p \wedge (\sim q)]$
d. $\sim(p \wedge q)$

11. The value of

$\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right) + \sin^{-1}\left(\frac{1}{3}\right)$ is equal to

a. $\frac{\pi}{6}$
b. $\frac{\pi}{2}$
c. $\frac{\pi}{4}$
d. $\frac{2\pi}{3}$

12. If the eccentricity of the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{5}{4}$ and $2x + 3y - 6 = 0$

Is a focal chord of the hyperbola, then the length of transverse axis is equal to

a. $\frac{12}{5}$
b. $\frac{24}{5}$
c. $\frac{6}{5}$
d. $\frac{5}{24}$

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KCET-2015 (Mathematics)

13. If $\vec{a} = i + 2j + 2k, \left|\vec{b}\right|$ = 5 and the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, then the area of the triangle formed by these two vectors as two sides is

a. $\frac{15}{2}$ b. 15

c. $\frac{15}{4}$ d. $\frac{15\sqrt{3}}{2}$

14. Let $\vec{a} = i - 2j + 3k$ if $\vec{b}$ is a vector such that $\vec{a}.\vec{b} = \left|\vec{b}\right|^2$ and $\left|\vec{a} - \vec{b}\right| = \sqrt{7}$, then $\left|\vec{b}\right| =$ _____

a. 7 b. 14

c. $\sqrt{7}$ d. 21

15. If direction cosines of a vector of magnitude 3 are $\frac{2}{3}, -\frac{9}{3}, \frac{2}{3}$ and a > 0, then vector is _____

a. 2i + j + 2k b. 2i – j + 2k

c. i –2j + 2 k d. i + 2j + 2k

16. Equation of line passing through the point (2, 3, 1) and parallel to the line of intersection of the plane x – 2y – z + 5 = 0 and x + y + 3z = 6 is

a. $\frac{x-2}{5} = \frac{y-3}{-4} = \frac{z-1}{3}$ b. $\frac{x-2}{-5} = \frac{y-3}{-4} = \frac{z-1}{3}$

c. $\frac{x-2}{5} = \frac{y-3}{4} = \frac{z-1}{3}$ d. $\frac{x-2}{4} = \frac{y-3}{3} = \frac{z-1}{2}$

17. Foot of perpendicular drawn from the origin to the plane 2x – 3y + 4z = 29 is _____

a. (5, –1, 4) b. (2, –3, 4)

c. (7, –1, 3) d. (5, –2, 3)

18. If two dice are thrown simultaneously, then the probability that the sum of the number which come up on the dice to be more than 5 is _____.

a. $\frac{5}{36}$ b. $\frac{1}{6}$

c. $\frac{5}{18}$ d. $\frac{13}{18}$

19. If $y = f(x^2 + 2)$ and f `(3) = 5, then $\frac{dy}{dx}$ at x = 1 is _____

a. *5* b. 25

c. 15 d. 10

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20. If $x = a\cos^3\theta$, $y = a\sin^3\theta$, then $1 + \left(\frac{dy}{dx}\right)^2$ is _____

a. $\tan\theta$ b. $\tan^2\theta$
c. $\sec^2\theta$ d. 1

21. Slope of Normal to the curve
$y = x^2 - \frac{1}{x^2}$ at $(-1, 0)$ is

a. $\frac{1}{4}$ b. $-\frac{1}{4}$
c. 4 d. -4

22. $\int \frac{1}{x^2(x^4+1)^{3/4}}\,dx$ is equal to _____

a. $\frac{-(1+x^4)^{1/4}}{x} + C$ b. $\frac{-(1+x^4)^{1/4}}{x^2} + C$
c. $\frac{-(1+x^4)^{1/4}}{2x} + C$ d. $\frac{-(1+x^4)^{3/4}}{x} + C$

23. If $f : R \to R$ is defined by $f(x) = \frac{x}{x^2+1}$, find $f(f(2))$

a. $\frac{1}{29}$ b. $\frac{10}{29}$
c. $\frac{29}{10}$ d. 29

24. Evaluate $\begin{vmatrix} \cos 15 & \sin 15 \\ \sin 75 & \cos 75 \end{vmatrix}$

a. 1 b. 0
c. 2 d. 3

25. A man takes a step forward with probability 0.4 and one step backward with probability 0.6. then the probability that at the end of eleven steps he is one step away from the starting point is

a. ${}^{11}C_5 \times (0.48)^5$ b. ${}^{11}C_6 \times (0.24)^5$
c. ${}^{11}C_5 \times (0.12)^5$ d. ${}^{11}C_6 \times (0.72)^6$

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26. $\int_0^{\pi/4} \log\left(\frac{\sin x + \cos x}{\cos x}\right) dx$

a. $\frac{\pi}{4}\log 2$
b. $\frac{\pi}{2}\log 2$
c. $\frac{\pi}{8}\log 2$
d. Log 2

27. Area bounded by $y = x^3$, $y = 8$ and $x = 0$ is _____
a. 2 sq. unit
b. 4 sq. unit
c. 12 sq. unit
d. 6 sq. unit

28. Let $\vec{a} = i + 2j + k, \vec{b} = i - j + k$ and $\vec{c} = i + j - k$, a vector in the plane $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$ is _____.
a. 3i + j – 3k
b. 4i + j – 4k
c. I+ j – 2k
d. 4i – j + 4k

29. The mean deviation from the data 3, 10, 10, 4, 7, 10, 5 :
a. 3
b. 2
c. 3. 75
d. 2.75

30. The probability distribution of x is

x	0	1	2	3
P(x)	0.2	k	k	2k

Find the value of k
a. 0.2
b. 0.3
c. 0.4
d. 0.1

31. If the function g(x) is defined by
$g(x) = \frac{x^{200}}{200} + \frac{x^{199}}{199} + \frac{x^{198}}{198} + ... + \frac{x^2}{2} + x + 5$
a. 1
b. 200
c. 100
d. 5

32. A box contains 6 red marbles numbers from 1 through 6 and 4 white marbles 12 through 15 Find the probability that a marble drawn 'at random' is white and odd numbered.
a. 5
b. $\frac{1}{5}$
c. 6
d. $\frac{1}{6}$

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KCET-2015 (Mathematics)

33. $\lim_{x\to 0}\frac{1-\cos x}{x^2}$ is ___

a. 2
b. 3
c. $\frac{1}{2}$
d. $\frac{1}{3}$

34. $f(x)=\begin{cases}3x-8 & \text{if } x\le 5\\ 2k & \text{if } x>5\end{cases}$ is continuous, find k.

a. $\frac{2}{7}$
b. $\frac{3}{7}$
c. $\frac{4}{7}$
d. $\frac{7}{2}$

35. If $f(x) = 2x^2$, find $\frac{f(3.8)-f(4)}{3.8-4}$ Choose the correct option:

a. 1.56
b. 156
c. 15.6
d. 0.156

36. If $x = ct$ and $y = \frac{c}{t}$, find $\frac{dy}{dx}$ at $t = 2$.

a. $\frac{1}{4}$
b. 4
c. $\frac{-1}{4}$
d. 0

37. A balloon which always remains spherical is being inflated by pumping in 10 cubic centimeters of gas per second. Find the rate at which the radius of the balloon is increasing when the radius is 15 cms

a. $\frac{1}{90\pi}\text{cm/sec}$
b. $\frac{1}{9\pi}\text{cm/sec}$
c. $\frac{1}{30\pi}\text{cm/sec}$
d. $\frac{1}{\pi}\text{cm/sec}$

38. $\int\frac{\sin^2 x}{1+\cos x}dx$

a. x + sinx + C
b. x – sinx + C
c. Sin x + C
d. Cos x + C

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39. $\int e^x\left(\frac{1+\sin x}{1+\cos x}\right)dx$

a. $e^x \tan\left(\frac{x}{2}\right)+C$ b. $\tan\left(\frac{x}{2}\right)+C$

c. $e^x + C$ d. $e^x \sin x + C$

40. If 1, w, w^2 are three cube roots of unity, then $(1-w+w^2)(1+w-w^2)$ is _____

a. 1 b. 2

c. 3 d. 4

41. Solve for x

$\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x, x>0$

a. $\sqrt{3}$ b. 1

c. –1 d. $\frac{1}{\sqrt{3}}$

42. The system of linear equation x + y + z = 6, x + 2y + 3z = 10 and x + 2y + az = b has no solutions when_____

a. a = 2 b ≠ 3 b. a = 3 b ≠ 10

c. b = 2 a = 3 d. b = 3 a ≠ 3

43. The value of tan (1°) + tan(89°) is _____

a. $\frac{1}{\sin(1^\circ)}$ b. $\frac{2}{\sin(2^\circ)}$

c. $\frac{2}{\sin(1^\circ)}$ d. $\frac{2}{\sin(2^\circ)}$

44. If $\frac{(x+1)^2}{x^3+x}=\frac{A}{x}+\frac{Bx+C}{x^2+1}$, then $\text{cosec}^{-1}\left(\frac{1}{A}\right)+\cot^{-1}\left(\frac{1}{B}\right)+\sec^{-1}C$ = _____

a. $\frac{5\pi}{6}$ b. 0

c. $\frac{5\pi}{6}$ d. $\frac{\pi}{2}$

45. The remainder obtained when 1! + 2! + 3! + ……+ 11! Is divided by 12 is _____

a. 9 b. 8

c. 7 d. 6

46. If $\alpha \le 2\sin^{-1}x + \cos^{-1}x \le \beta$, then

a. $\alpha=\frac{-\pi}{2}\ \beta=\frac{\pi}{2}$ b. $\alpha=\frac{-\pi}{2}\ \beta=\frac{3\pi}{2}$

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c. $\alpha = 0 \ \beta = \pi$ d. $\alpha = 0 \ \beta = 2\pi$

47. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then A^2 equal to _____

a. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ d. $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

48. The function f(x) = [x], where [x] denotes greatest integer function is continuous at _____

a. 4 b. –2

c. 1 d. 1.5

49. If $y = \log\left(\frac{1-x^2}{1+x^2}\right)$, then $\frac{dy}{dx}$ is equal to _____

a. $\frac{-4x}{1-x^4}$ b. Stem height

c. $\frac{1}{4-x^4}$ d. $\frac{-4x^3}{1-x^4}$

50. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 = 2$

a. Touch each other b. Cut at right angle

c. Cut at angle $\frac{\pi}{3}$ d. Cut at angle $\frac{\pi}{4}$

51. If x is real, then the minimum value of $x^2 - 8x + 17$ is _____

a. 1 b. 2

c. 3 d. 4

52. $\int_{-\pi/4}^{\pi/4} \frac{dx}{1+\cos 2x}$ is equal to

a. 2 b. 1

c. 4 d. 0

53. The order of differential equation of all circles of given radius 'a' is _____

a. 4 b. 2

c. 1 d. 3

54. The solution of differential equation

$x\frac{dy}{dx} + 2y = x^2$ is _____

a. $y = \frac{x^2 + C}{4x^2}$ b. $y = \frac{x^2}{4}$

c. $y = \frac{x^4 + C}{x^2}$ d. $y = \frac{x^4 + C}{4x^2}$

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KCET-2015 (Mathematics)

55. If $\sin x + \sin y = \frac{1}{2}$ and $\cos x + \cos y = 1$, then $\tan(x + y) =$ _____

a. $\frac{8}{3}$ b. $-\frac{3}{4}$

c. $\frac{-8}{3}$ d. $\frac{4}{3}$

56. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 27$, then $\alpha =$ _____

a. ± 1 b. ± 2

c. $\pm\sqrt{7}$ d. $\pm\sqrt{5}$

57. If $A = \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix}$ and $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix}$, then $\frac{dQ}{dx} =$ _____

a. 3p +1 b. 1–3p

c. –3p d. 3p

58. A line passes through (2, 2) and is perpendicular in the line 3x + y = 3 its y-intercept is ___

a. $\frac{1}{3}$ b. $\frac{2}{3}$

c. $\frac{4}{3}$ d. 1

59. Let $f : R \to R$ be defined by $f(x) = \frac{1}{x} \forall x \in R$, then f is _____

a. One-*one* b. onto

c. bijective d. f is not defined

60. The solution set of the inequation $\frac{x^2 + 6x - 7}{|x + 4|} < 0$ is _____

a. (–7, 1) b. (–7, –4)

c. $(-7, -4) \cup (-4, 1)$ d. $(-7, -4) \cup (4, 1)$

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KCET-2015 (Mathematics)

ANSWER KEYS

* G – Indicates one Grace mark awarded for the question number.

1. (a)	2. (c)	3. (d)	4. (c)	5. (a)	6. (a)	7. (d)	8. (d)	9. (a)	10. (G)
11. (b)	12. (b)	13. (c)	14. (c)	15. (G)	16. (b)	17. (b)	18. (d)	19. (d)	20. (c)
21. (a)	22. (a)	23. (b)	24. (b)	25. (b)	26. (c)	27. (c)	28. (d)	29. (d)	30. (a)
31. (a)	32. (b)	33. (c)	34. (d)	35. (c)	36. (c)	37. (a)	38. (b)	39. (d)	40. (d)
41. (d)	42. (b)	43. (b)	44. (b)	45. (a)	46. (c)	47. (c)	48. (d)	49. (a)	50. (b)
51. (a)	52. (a)	53. (b)	54. (d)	55. (d)	56. (c)	57. (d)	58. (c)	59. (d)	60. (c)

KCET-2015 (Mathematics)

Solution

1. (a)

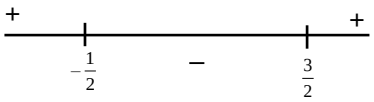
$g(x) = \sqrt{3+4x-4x^2}$

Domain of $g(x) \Rightarrow \quad 3+4x-4x^2 \geq 0$

$4x^2 - 4x - 3 \leq 0$

$(2x-1)(2x-3) \leq 0$

$x = -\frac{1}{2}, 3/2$



$X \in \left[-\frac{1}{2}, \frac{3}{2}\right]$

$f(x) = \frac{1}{2} - \tan\left(\frac{\pi x}{2}\right)$ $\quad \{\because -1 < x < 1\}$

Domain of $f(x) \Rightarrow (-1, 1)$

$\Rightarrow$ Domain of $(f + g)(x)$

$\Rightarrow$ Domain of $f(x) \cap$ Domain of $g(x)$

$\Rightarrow (-1, 1) \cap \left[-\frac{1}{2}, \frac{3}{2}\right]$

$\Rightarrow \left[-\frac{1}{2}, -1\right)$

So Domain of $(f + g) \Rightarrow \left[-\frac{1}{2}, 1\right)$

2. (c)

Consider the set A $\{-1, 1\}$

Here -1 and 1 are the roots of the equation $x^2 - 1 = 0$

So set A in set builder form can be written as

$A = \{x : x \text{ is a root of equation } x^2 - 1 = 0\}$

3. (d)

Given that

$a \oplus b = a^2 + b^2$

We have

$(2 \oplus 3) \oplus 4 = (2^2 + 3^2) \oplus 4$

$= (4 + 9) \oplus 4$

$= 13 \oplus 4$ {Using given condition}

$= 13^2 + 4^2$

$= 169 + 16$

$= 185$

4. (c)

$Z = \dfrac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}$

$|Z| = \left|\dfrac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}\right|$

$|Z| = \left|\dfrac{|\sqrt{3}+i|^3|3i+4|^2}{|8+6i|^2}\right|$

{using the identify $|a + ib| = \sqrt{a^2+b^2}$

$|Z| = \dfrac{(3+1)^{\frac{3}{2}}(9+16)^{\frac{2}{2}}}{(64+36)^{\frac{2}{2}}}$

$|Z| = \dfrac{(8)(25)}{(100)}$

$|Z| = 2$

5. (a)

Given Q. E $\Rightarrow x^2 - ax + b^2 =$ 

Sum of root $\Rightarrow \alpha + \beta = a$

Product of root $\Rightarrow \alpha\beta = b^2$

Now $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a)^2 - 2b^2$

6. (a)

2nd term of G.P $\Rightarrow ar = 24 \Rightarrow a = \dfrac{24}{r}$ .....(1)

5th term of G.P. $\Rightarrow ar^4 = 3$ .....(2)

Eq. (1) put in Eq (2)

$\Rightarrow \dfrac{24}{r} . r^4 = 3$

$\Rightarrow 8r^3 = 1$

$r = \dfrac{1}{2}$

So $a = \dfrac{24}{1/2} = 48$

Sum of first six terms is

$\Rightarrow a + ar + ar^2 + ar^3 + ar^4 + ar^5$

$\Rightarrow 48 + 24 + 12 + 6 + 3 + \dfrac{3}{2}$

$\Rightarrow \dfrac{189}{2}$

7. (d)
n = 10 $\Rightarrow$ even

middle terms = $\left(\frac{10}{2}+1\right)^{th}$ = 6th term

$T_6 = T_{5+1} = {}^{10}C_5\left(\frac{10}{x}\right)^5\left(\frac{x}{10}\right)^5 = {}^{10}C_5$

8. (d)
Area of triangle whose vertices are

$\left(\frac{x_1}{a},\frac{y_1}{a}\right),\left(\frac{x_2}{b},\frac{y_2}{b}\right),\left(\frac{x_3}{c},\frac{y_3}{c}\right)$ is

$$\Delta = \frac{1}{2}\begin{vmatrix} \frac{x_1}{a} & \frac{y_1}{a} & 1 \\ \frac{x_2}{b} & \frac{y_2}{b} & 1 \\ \frac{x_3}{c} & \frac{y_3}{c} & 1 \end{vmatrix}$$

On multiplying R_1, R_2 and R_3 by a. b . c respectively we get

$$\Delta = \frac{1}{2abc}\begin{vmatrix} x_1 & y_1 & a \\ x_2 & y_2 & b \\ x_3 & y_3 & c \end{vmatrix}$$

On multiplying C_3 by 2 we get

$$\Delta = \frac{1}{4abc}\begin{vmatrix} x_1 & y_1 & 2a \\ x_2 & y_2 & 2b \\ x_3 & y_3 & 2c \end{vmatrix}$$

On applying $C_1 \leftrightarrow C_3$, we get

$$\Delta = \frac{1}{4abc}\begin{vmatrix} 2a & y_1 & x_1 \\ 2b & y_2 & x_2 \\ 2c & y_3 & x_3 \end{vmatrix}$$

Apply $C_2 \leftrightarrow C_3$

$$\Delta = \frac{1}{4abc}\begin{vmatrix} 2a & x_1 & y_1 \\ 2b & x_2 & y_2 \\ 2c & x_3 & y_3 \end{vmatrix}$$

$= \frac{1}{4abc}\left(\frac{abc}{2}\right)$

$= \frac{1}{8}$

9. (a)
Equation of line passing through the coordinate (0, 14) and (9, 14) is
y = 14
Consider the equation of the line joining (0, 14) and (5, 0) applying intercept form gives us

$\Rightarrow \quad \frac{x}{5}+\frac{y}{14}=1$

$\Rightarrow$ 14x + 5y = 70
The equation of the line joining (5, 0) and (19, 14) is given by

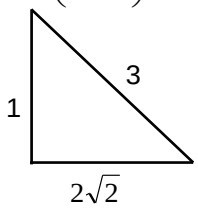
$\Rightarrow \quad \frac{y-14}{x-19}=\frac{y-0}{x-5}$

$\Rightarrow$ xy – 14x – 5y + 70 = xy – 19y
x – y = 5
Consider the figure and observe the shaded region the inequalities can be written as
$y \le 14$
$14x + 5y \ge 70$
$x - y \le 5$.

10. (G) Bonus

11. (b)

$\Rightarrow \quad \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right) + \sin^{-1}\left(\frac{1}{3}\right)$



$\Rightarrow \quad \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right) + \cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

$= \quad \frac{\pi}{2} \qquad \left\{\because \sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}\right\}$

12 (b)
Given hyperbola has focus (ae, 0) which will lie on 2x + 3y – 6 = 0 as it is focal chord

$\therefore$ 2ae – 6 = 0
2ae = 6

ae =3 $\qquad \left[\because e=\frac{5}{4}\right]$

$a\left(\frac{5}{4}\right) = 3$; $a = \frac{12}{5}$

$\therefore$ Length of transverse axis $= 2a$

$= 2 \times \frac{12}{5} = \frac{24}{5}$

13 (c)

$a = a = \hat{i} + 2\hat{j} + 2\hat{k}$

$|a| = \sqrt{(1)^2 + (2)^2 + (2)^2}$

$|a| = \sqrt{1+4+4}$

$|a| = 3$

$\therefore$ Required area $= \frac{1}{2}|a \times b| = \frac{1}{2} \times a \times b \times \sin\theta$

$= \frac{1}{2} \times 3 \times 5 \times \sin\frac{\pi}{6} \qquad \left\{\because \theta = \frac{\pi}{6}\right\}$

$= \frac{15}{4}$

14 (c)

$|\vec{a} - \vec{b}| = \sqrt{7}$

Squaring both side

$\Rightarrow \quad |\vec{a}|^2 + |\vec{b}|^2 - 2a.b = 7 \qquad \left\{\because \vec{a}.\vec{b} = |\vec{b}|^2\right\}$

$\Rightarrow \quad \left[\sqrt{(1)^2 + (-2)^2 + (3)^2}\right]^2 + |\vec{b}|^2 - 2|\vec{b}|^2 = 7$

$\Rightarrow \quad 14 - |\vec{b}|^2 = 7$

$|\vec{b}|^2 = 7$

$|\vec{b}| = \sqrt{7}$

15 (G) Bonus

16 (b)

Consider the planes

$X - 2y - Z + 5 = 0$ and $x + y + 3z = 6$

Let the DR's of the line passing through the intersection of the given plane be a, b and C

$a - 2b - C = 0$ and $a + b + 3c = 0$

Thus

$$\frac{a}{-6+1}=\frac{-6}{3+1}=\frac{C}{1+2}$$

$$\frac{a}{-5}=\frac{-b}{4}=\frac{c}{3}$$

Now the required line passes through (2, 3, 1) and is parallel to the above line.
The equation of the required line is,

$$\frac{x-2}{-5}=\frac{y-3}{-4}=\frac{z-1}{3}.$$

17 (b)

Consider the equation of plane,

$2x - 3y + 4z = 29$

The DR's of the line perpendicular to line form origin is 2, – 3 and 4

The equation of the line will be

$$\frac{x-0}{2}=\frac{y-0}{-3}=\frac{z-0}{4}=\lambda$$

So the coordinates of the foot is $(2\lambda, -3\lambda, 4\lambda)$

Since the foot lies on the plane, hence

$4\lambda + 9\lambda + 16\lambda = 29$

$29\lambda = 29$

$29\lambda = 1$

Thus, the coordinate of the foot are (2, –3, 4)

18 (d)

Let E be the event of getting sum more than 5.

$$E=\begin{Bmatrix}(1,5)\ (1,6)\ (2,4)\ (2,5)\ (2,6)\ (3,3)(3,4)(3,\ 5)(3,6)\\(4,2)\ (4,3)\ (4,4)\ (4,5)\ (4,6)\ (5,1)(5,2)(5,\ 3)(5,4)\\(5,5)\ (5,6)\ (6,1)\ (6,2)\ (6,3)\ (6,4)(6,5)(6,\ 6)\end{Bmatrix}$$

The probability of getting sum more than 5 is calculate as

$$P(E) = \frac{26}{36} = \frac{13}{18}$$

19 (d)

$y = f(x^2 + 2)$

Differentiate w.r.t x

$\frac{dy}{dx} = f`(x^2 + 2).\ 2x$

$\left.\frac{dy}{dx}\right|_{x=1} = f`(1 + 2) \times 2(1)$

$= f`(3) \times 2$

$= 2(5) = 10.$

20 (c)

Consider the expression.

$X = a\cos^3\theta$

Differentiate w.r.t θ

$\frac{dx}{d\theta} = -3a\cos^2\theta \cdot \sin\theta$ .....(1)

and $y = a\sin^3\theta$

Differentiate w.r.t θ

$\frac{dy}{d\theta} = 3a\sin^2\theta \cdot \cos\theta$ .....(2)

Divide equation (2) by (1)

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta}$$

$$\frac{dy}{dx} = -\frac{\sin\theta}{\cos\theta} = -\tan\theta$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)^2 + 1 = (-\tan\theta)^2 + 1$$

$= 1 + \tan^2\theta = \sec^2\theta$

21 (a)

$y = x^2 - \frac{1}{x^2}$

Differentiate w.r.t x

$\frac{dy}{dx} = 2x + \frac{2}{x^3}$

$$\left.\frac{dy}{dx}\right|_{x=-1} = 2(-1) + \frac{2}{(-1)^3}$$

$= -2 - 2 \qquad = -4$

Slope = $\frac{-1}{\frac{dy}{dx}} = \frac{-1}{-4} = \frac{1}{4}$

22 (a)

Let

$$I = \int \frac{1}{x^2(x^4+1)^{3/4}}\,dx$$

$$I = \int \frac{1}{x^2.x^3\left(1+\frac{1}{x^4}\right)^{3/4}}\,dx$$

$$I = \int \frac{1}{x^5\left(1+\frac{1}{x^4}\right)^{3/4}}\,dx$$

Let $1+\frac{1}{x^4} = t$; $dx = \frac{-x^5}{4}dt$

$$I = \frac{-1}{4}\int \frac{1}{(t)^{3/4}}\,dt$$

$$I = \frac{-1}{4}.\frac{t^{1/4}}{1/4} + C$$

$$I = -\left(1+\frac{1}{x^4}\right) + C$$

$$I = \frac{-(x^4+1)^{\frac{1}{4}}}{x} + C$$

23. (b)

$$f(x) = \frac{x}{x^2+1} \Rightarrow f(f(x)) = \frac{f(x)}{[f(x)]^2+1} \Rightarrow f(f(x)) = \frac{\frac{x}{x^2+1}}{\left(\frac{x}{x^2+1}\right)^2+1}$$

Put $x = 2 \Rightarrow f\left(f(2)\right) = \dfrac{\dfrac{2}{4+1}}{\left(\dfrac{2}{4+1}\right)^2+1}$

$= \dfrac{\dfrac{2}{5}}{\dfrac{4}{25}+1} = \dfrac{2}{5}\times\dfrac{25}{29} = \dfrac{10}{29}$

24. (b)

$let\ \mathrm{D} = \begin{vmatrix} \cos\ 15° & \sin\ 15° \\ \cos\ 75° & \cos\ 75° \end{vmatrix}$

= cos 15° cos 75° – sin 15° sin 75°
= cos (15 + 75)
= cos 90°
= 0

25. (b)
One step away means a step forward or a step backward $\Rightarrow$ either he took 6 forward steps and 5 backward steps or 6 backward steps and 5 forward steps
$\Rightarrow$ required probability
$= {}^{11}C_6(0.4)^6(0.6)^5 + {}^{11}C_5\ (0.4)^5(0.6)^6 = {}^{11}C_5(0.4)^5(0.6)^5[0.4+0.6]$
$= {}^{11}C_5(0.24)^5 = {}^{11}C_6(0.24)^5$

26. (c)
Let

$$I = \int_0^{\frac{\pi}{4}} \log\left(\frac{\sin x + \cos x}{\cos x}\right)dx$$

$$I = \int_0^{\frac{\pi}{4}} \log(1+\tan\ x)\,dx \qquad .....(1)$$

Using property

$$I = \int_0^{\frac{\pi}{4}} \log\left(1+\tan\ \left(\frac{\pi}{4}-x\right)\right)dx$$

$$I = \int_0^{\frac{\pi}{4}} \log\left(\frac{2}{1+\tan\ x}\right)dx \qquad .....(2)$$

Add eq. (1) and (2)

$$2I = \int_0^{\frac{\pi}{4}} \left[\log(1+\tan x) + \log\left(\frac{2}{1+\tan x}\right) \right] dx$$

Using log property

$$2I = \int_0^{\frac{\pi}{4}} \log 2 \, dx$$

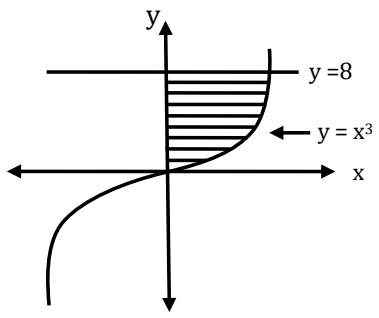
$$2I = \log 2 \left[x \right]_0^{\frac{\pi}{4}}$$

$$2I = \log 2 \left[\frac{\pi}{4} - 0 \right]$$

$$I = \frac{\pi}{8} \log 2$$

27. (c)

Consider the graph of $y = x^3$ and $y = 8$ on the same plane.



The required area is the shaded region.

$$A = \int_0^8 y^{\frac{1}{3}} dy$$

$$A = \frac{3}{4} \left[y^{4/3} \right]_0^8$$

$$A = \frac{3}{4} [16]$$

$$A = 12 \text{ sq. units}$$

28. (d)

Given vectors,

$\vec{a} = \hat{i} + 2\hat{i} + k$ ،

$\vec{b} = \hat{i} - \hat{i} + k$

$\vec{c} = \hat{i} + j - k$

The vector r in the plane a and b is calculated as

$\vec{r} = \vec{a} + \lambda \vec{b}$

$\vec{r} = (1+\lambda)\hat{i} + (2-\lambda)\hat{i} + (1+\lambda)k$ .....(1)

Projection = $\frac{\vec{r}.\vec{c}}{|\vec{c}|}$

$$\left|\frac{1}{\sqrt{3}}\right| = \frac{\left[(1+\lambda)\hat{i}+(2-\lambda)\hat{i}+(1+\lambda)k\right].\left[\hat{i}+\hat{i}+k\right]}{\sqrt{1+1+1}}$$

$$\left|\frac{1}{\sqrt{3}}\right| = \frac{1+\lambda+2-\lambda-1-\lambda}{\sqrt{3}}$$

$$2-\lambda = \pm 1$$

$$2-\lambda = 1 \quad | \quad 2-\lambda = -1$$

$$\lambda = 1 \qquad \lambda = 3$$

Value of λ put in eq. (1)

$\vec{r} = 2\hat{i}+\hat{i}+3k,\ 4\hat{i}-\hat{i}+4k$

29. (d)

$$Mean = \frac{3+4+5+7+10+10+10}{7}$$

$$= \frac{49}{7}$$

$$= 7$$

$$Median = \left(\frac{7+1}{2}\right)^{th} \text{ observation}$$

$= 4^{th}$ observation

$Median = 7$

$$Mean \text{ deviation} = \frac{1}{7}\left[|7-3|+|7-4|+|7-5|+|7-7|+|7-10|+|7-10|+|7-10|\right]$$

$$= \frac{1}{7}\left[4+3+2+0+3+3+3\right]$$

$$= \frac{18}{7} = 2.57$$

30. (a)

$$\sum P(x) = 1$$

$$0.2+k+k+2k = 1$$

$$0.2+4k = 1$$

$$4k = 0.8$$

$$k = 0.2$$

31. (a)

$$g(x) = \frac{x^{200}}{200}+\frac{x^{199}}{199}+\frac{x^{198}}{198}+.....+\frac{x^2}{2}+x+5$$

Differentiate w.r.t x

$$g'(x)=\frac{200x^{199}}{200}+\frac{199x^{198}}{199}+\frac{198x^{197}}{197}+.....+\frac{2x}{2}+1$$
$$=x^{199}+x^{198}+x^{197}+.....+x+1$$
$put\ x=0$
$$g'(0)=0+0+0+.....+0+1$$
$$g'(0)=1$$

32. (b)
The total number of marbles are n(S) = 6 + 4 = 10
The number of marbles that are white and odd numbered is n(E) = 2
$$\Pr obability=\frac{n(E)}{n(S)}=\frac{2}{10}=\frac{1}{5}$$

33. (c)
$$\Rightarrow \lim_{x\to 0}\frac{1-\cos x}{x^2} \qquad \left\{\frac{0}{0}\ \text{form}\right\}$$
Using L' Hospital rule
$$\Rightarrow \lim_{x\to 0}\frac{\sin x}{2x}$$
$$\Rightarrow \frac{1}{2}\lim_{x\to 0}\frac{\sin x}{x}$$
$$=\frac{1}{2}(1)$$
$$=\frac{1}{2}$$

34. (d)
$$f(x)=\begin{cases}3x-8 & :\ x\le 5\\ 2k & :\ x>5\end{cases}$$
$$LHL\Rightarrow \lim_{x\to 5^-} f(x)=\lim_{x\to 5^-}(3x-8)$$
$$= 3(5) - 8$$
$$= 7 \qquad .....(1)$$
$$RHL\Rightarrow \lim_{x\to 5^+} f(x)=\lim_{x\to 5^+}2k=2k \qquad .....(2)$$
Value of function $\Rightarrow$ f(5) = 3 × 5 – 8 = 7
$\because$ $f(x)$ is continuous at $x = 5$
$\therefore$ LHL = RHL = f (5)
$$7=2k=7$$
$$\boxed{k=\frac{7}{2}}$$

35. (c)

given that $f(x) = 2x^2$

Now

$$\frac{f(3.8) - f(4)}{3.8 - 4} = \frac{2(3.8)^2 - 2(4)^2}{3.8 - 4}$$

$$= \frac{28.88 - 32}{-0.2}$$

$$= \frac{-3.12}{-0.2}$$

$$= 15.6$$

36. (c)

$x = ct$

differentiate w.r.t t

$\frac{dx}{dt} = c$ .....(1)

and

$y = \frac{c}{t}$

differentiate w.r.t t

$\frac{dy}{dt} = -\frac{c}{t^2}$ .....(2)

divide equation (2) by (1)

$$\frac{dy}{dt} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-\frac{c}{t^2}}{c} = -\frac{1}{t^2}$$

$$\left.\frac{dy}{dx}\right|_{t=2} = \frac{-1}{(2)^2} = -\frac{1}{4}$$

37. (a)

The rate of pumping the gas in the balloon is $\frac{dv}{dt} = 10cm^3/\sec$

The volume of the spherical balloon is given by $v = \frac{4}{3}\pi r^3$ differentiate w.r.t r

$$\frac{dv}{dr} = \frac{4}{3}\pi(3r^2)$$

$$= 4\pi r^2$$

$$\left.\frac{dv}{dr}\right|_{r=15} = 4\pi(15)^2 = 900\pi$$

Now,

$$\frac{dv}{dt} = \frac{dv}{dr} \times \frac{dr}{dt}$$

$$10 = 900\pi \times \left(\frac{dr}{dt}\right)$$

$$\frac{dr}{dt} = \frac{1}{90}\pi \text{ cm/sec}$$

38. (b)

$$Let\ I = \int \frac{\sin^2 x}{1+\cos x}dx$$

$$I = \int \frac{\left(1-\cos^2 x\right)}{1+\cos x}dx$$

$$I = \int \frac{(1-\cos x)(1+\cos x)}{(1+\cos x)}dx$$

$$I = \int (1-\cos x)dx$$

$$I = x - \sin x + c$$

39. (d)

$$Let\ I = \int e^x \left(\frac{1+\sin x}{1+\cos x}\right)dx$$

$$I = \int e^x \left(\frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{1+\cos^2\frac{x}{2}-1}\right)dx$$

$$\left\{\begin{array}{c}\because\ \sin 2A = 2\sin A\cos A \\ \cos 2A = 2\cos^2 A - 1\end{array}\right\}$$

$$I = \int e^x \left(\frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right)dx$$

$$I = \int e^x \left(\frac{1}{2\cos^2 \frac{x}{2}} + \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2 \frac{x}{2}} \right) dx$$

$$I = \int e^x \left(\frac{1}{2}\sec^2 \frac{x}{2} + \tan\frac{x}{2} \right) dx$$

$$I = e^x \tan\frac{x}{2} + c$$

40. (d)

If 1, w, w^2 are three cube roots of unity,

then $1 + w + w^2 = 0$ .....(1)

Hence

$(1-w + w^2)(1 + w - w^2) = (1 + w + w^2 - 2w)(1+ w + w^2 - 2w^2)$

$= (0 - 2w)(0 - 2w^2)$ {from eq. (1)}

$= (- 2w)(- 2w^2)$

$= 4w^3$

$= 4(1)$

$= 4$

41. (d)

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1} x$$

$$\Rightarrow \tan^{-1}(1) - \tan^{-1} x = \frac{1}{2}\tan^{-1} x \Rightarrow \tan^{-1} x + \frac{1}{2}\tan^{-1} x = \tan^{-1}(1)$$

$$\Rightarrow \frac{3}{2}\tan^{-1} x = \tan^{-1}(1) \Rightarrow \frac{3}{2}\tan^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} x = \frac{\pi}{6} \Rightarrow x = \tan\frac{\pi}{6}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}}$$

42. (b)

Consider the given system of equations

$x + y + z = 6$ .....(1)

$x + 2y + 3z = 10$ .....(2)

$x + 2y + az = 6$ .....(3)

$\because$ condition of no solution, $\Delta = 0$

So $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & a \end{vmatrix} = 0$

$\Rightarrow 1(2a-6)-1(a-3)+1(2-2)=0$

$\Rightarrow 2a-6-a+3=0$

$\Rightarrow a=3$

Now, b ≠ 10 then equations (2) and (3) will become identical, which will have infinite solutions.
The condition is satisfied for the option a = 3 and b ≠ 10.

43. (b)

$\Rightarrow \tan(1°)+\tan(89°)$

$= \tan(1°)+\tan(90-1°)$

$= \tan(1°)+\cot(1°) = \dfrac{\sin 1°}{\cos\ 1°}+\dfrac{\cos 1°}{\sin\ 1°}$

$= \dfrac{\sin^2(1°)+\cos^2(1°)}{\sin\ (1°)\cos(1°)} = \dfrac{1}{\sin(1°)\cos(1°)}$

$= \dfrac{2}{2\sin(1°)\cos(1°)} = \dfrac{2}{\sin(2°)}$

44. (b)

Consider the expression

$\text{cosec}^{-1}\left(\dfrac{1}{A}\right)+\cot^{-1}\left(\dfrac{1}{B}\right)+\sec^{-1}\text{c} \qquad .....(1)$

Also

$\Rightarrow \dfrac{x^2+2x+1}{\text{x}\left(x^2+1\right)} = \dfrac{A}{x}+\dfrac{Bx+C}{x^2+1}$

$\Rightarrow \dfrac{x^2+1}{\text{x}\left(x^2+1\right)}+\dfrac{2x}{x\left(x^2+1\right)} = \dfrac{A}{x}+\dfrac{Bx+C}{x^2+1}$

$\Rightarrow \dfrac{1}{x}+\dfrac{2}{\left(x^2+1\right)} = \dfrac{A}{x}+\dfrac{Bx+C}{x^2+1}$

Compare the like terms of the equation

A = 1

B = 0

C = 2

Value of A, B, C put in equation (1)

$\Rightarrow \text{cosec}^{-1}\left(\dfrac{1}{1}\right)+\cot^{-1}\left(\dfrac{1}{0}\right)+\sec^{-1}(2) \Rightarrow \dfrac{\pi}{2}+\cot^{-1}(\infty)+\dfrac{\pi}{3}$

$= \dfrac{5\pi}{6}+0 = \dfrac{5\pi}{6}$

45. (a)

Given that :1! 2!+3!+4!+----- +11!

=(1!+2!+3!)+(4!+5!+6!+-----+11!)

=(1+2+6) + (4!+5!+6!+----+11!)

=9+(4!+5!+6!+----+11!)

4!, 5!, 6! ----- 11! Are divisible by 12 so, when you will divide given expression by 12 the remainder be 9.

46. (c)

For the sine function

$$-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2}$$

$$0 \le \frac{\pi}{2} + \sin^{-1} x \le \pi$$

$$0 \le \sin^{-1} x + \cos^{-1} x + \sin^{-1} x \le \pi \left\{ \because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right\}$$

$$0 \le 2\sin^{-1} x + \cos^{-1} x \le \pi$$

Compare the given expression

$\alpha = 0$

$\beta = \pi$

47. (c)

$$A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$A^2 = A.A$

$$=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$=\begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix}$$

$$=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

48. (d)

Consider the function

F(x) = [x]

Here, [x] denotes the greatest integer function.

[x] is always continuous at non integer value of x.

Thus, the function will be continuous at x=1.5

49. (a)

$Y = \log\left(\frac{1-x^2}{1+x^2}\right)$

using log property

$y=\log(1-x^2)-\log(1+x^2)$

differentiate w.r.t x

$\frac{dy}{dx} = \frac{1}{(1-x^2)}(-2x) - \frac{1}{(1+x^2)}(2x)$

$=(-2x)\left[\frac{1}{1-x^2}+\frac{1}{1+x^2}\right]$

$=(-2x)\left[\frac{1+x^2+1-x^2}{(1-x^2)(1+x^2)}\right]\left\{\because (A-B)(A+B) = A^2-B^2\right\}$

$=\frac{-4x}{(1-x^4)}$

50. (b)

first curve equation

$x^3-3xy^2+2=0$

differentiate w.r.t. x

$3x^2-3(y^2+2xy\,y')=0$

$3x^2-3y^2-6xy\,y')=0$

$y'=\frac{3x^2-3y^2}{6xy}=m_1$

second curve equation

$3x^2y-y^3=2$

Differentiate w.r.t x

$\Rightarrow 3(2xy + x^2y')-3y^2y' = 0$

$\Rightarrow 6xy + 3x^2y'-3y^2y' = 0$

$\Rightarrow y^1 = \frac{-6xy}{3x^2-3y^2}=m_2$

Calculate the product of slope

$m_1 \times m_2 = \left(\frac{-6xy}{3x^2-3y^2}\right)\left(\frac{3x^2-3y^2}{6xy}\right)$

$m_1 \times m_2 = -1$

The product is -1. So the curves intersect at each other at right angle.

51. (a)

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Let
$y=x^2-8x+17$
$y=(x^2-8x+16)+1$
$y=(x-4)^2+1$
so, $y\geq 1$ for all real values of x as $(x-4)^2\geq 0$
Thus, the minimum value of y is 1

52. (a)
Let

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{1+\cos 2x}$$

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{1+2\cos^2 x-1}$$

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{1}{2}\sec^2 x$$

Let
$f(x) = \sec^2 x$
$f(-x)=\sec^2(-x)$
$f(-x)=\sec^2 x$
$f(-x)=f(x)$
so f(x) is an even function
Therefore,

$$I=2\int_0^{\pi/4}\frac{1}{2}\sec^2 x\,dx$$

$$I=\int_0^{\pi/4}\sec^2 x\,dx$$

$$I=\left[\tan x\right]_0^{\pi/4}$$

$$I=\tan\frac{\pi}{4}-\tan 0$$

I=1-0
I=1

53. (b)
Let the centre of circle be (h,k) and radius be r.
The equation of circle can now be written as,
$(x-h)^2+(y-k)^2 = x^2$ .....(1)
Hence the differential equation is of order 2.

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54. (d)

$x\frac{dy}{dx}+2y=x^2$

$\frac{dy}{dx}+\frac{2}{x}.y=x$

The above equation is a linear equation in terms of y.

Therefore,

I.F = $e^{\int\frac{2}{x}dx}$

= $e^{2\log x}$

= $e^{\log x^2}$

= x^2

The required solution is

$y.x^2=\int(x.x^2)dx+c_1$

$yx^2=\frac{x^4}{4}+c_1$

$y=\frac{x^4+4c_1}{4x^2}$

$y=\frac{x^4+c}{4x^2}$

55. (d)

$\sin x+\sin y=\frac{1}{2}$

$2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)=\frac{1}{2}$ ...(1)

and

$\cos x+\cos y=1$

$2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)=1$ ...(2)

Divide equation (1) and (2)

$\Rightarrow\frac{2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}{2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}=\frac{1}{2}$

$\Rightarrow\tan\left(\frac{x+y}{2}\right)=\frac{1}{2}$

Now,

$\tan(x+y)=\frac{2\tan\left(\frac{x+y}{2}\right)}{1-\tan^2\left(\frac{x+y}{2}\right)}$

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$= \dfrac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2}$

$= \dfrac{4}{3}$

56. (c)

Given

$A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$

$|A| = \begin{vmatrix} \alpha & 2 \\ 2 & \alpha \end{vmatrix}$

$|A| = \alpha^2 - 4$

Now,

$|A^3| = 27$

$\left(\alpha^2 - 4\right)^3 = 3^3$

$\alpha^2 - 4 = 3$

$\alpha^2 = 7$

So,

$\alpha = \pm\sqrt{7}$

57. (d)

$P = \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix}$

$|P| = x^2-1$

and

$\theta = \begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix}$

$\theta = x(x^2-1)-1(x-1)+1(1-x)$

$\theta = x^3-x-x+1+1-x$

$\theta = x^3-3x+2$

differentiate w.r.t x

$\dfrac{d\theta}{dx} = 3x^2-3$

$= 3(x^2-1)$

$= 3P$

58. (c)

The equation of line is $3x+y = 3$

slope of the line $\Rightarrow 3+\dfrac{dy}{dx} = 0$

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$\Rightarrow \frac{dy}{dx}$ = -3 = m

slope of the perpendicular line $\Rightarrow \frac{-1}{m} = \frac{-1}{-3} = \frac{1}{3}$

The equation of line passing through (2,2) and perpendicular to given line

$\Rightarrow$ (y-2) = $\frac{1}{3}$(x-2)

3y-x = 4

$\frac{3y}{4} - \frac{x}{4} = 1$

Thus, the y-intercept is $\frac{4}{3}$

59. (d)

consider the function

f(x) = $\frac{1}{x}$ $\quad \forall\ x \in R$

substitute 0 for x in the above expression

f(x) = $\frac{1}{0}$

= undefined

Thus, the function f(x) is not defined

60. (c)

$\frac{x^2+6x-7}{|x+4|} < 0$

for the above expression

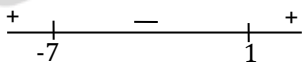
|x+4|>0

and

x^2+6x-7<0 $\quad \{ \because\ x \neq -4 \}$

(x+7)(x-1)<0

Diagram



So,

x← (-7, 1) – {-4} $\quad \{ \because\ x \neq -4 \}$

x← (-7, -4) ∪ (-4, 1)

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